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EDUCATION

Nanjing University of Information Science and Technology, China | 2022 - 2024

MSc in Artificial Intelligence

University of Plymouth, UK | 2017 - 2020

BSc (Hons) Software Engineering

- *Awarded - First Class Honors*
- *Aggregate Final Mark: 77.19 (GPA-4.0)*

ESOFT Metro Campus, Sri Lanka | 2018 - 2019

Diploma in Web Engineering

- *Offered by Pearson*

St. Nicholas International College, Sri Lanka | 2016

Cambridge GCE AS Level

St. Nicholas International College, Sri Lanka | 2014

Cambridge GCE Ordinary Level

WORK EXPERIENCE

Arenberg AG | Senior Software Engineer – Backend & AI Systems | Jan 2026 – Present

- Contributing to the development of Arenberg, a financial intelligence and trading/market monitoring platform that processes market news, economic indicators, financial events, and real-time market data to provide data-driven and AI-assisted insights.
- Designing and implementing scalable backend services and data pipelines using Python, FastAPI, PostgreSQL, MongoDB, Redis, and Google BigQuery.
- Developing Xyron, the trading and market monitoring component of the platform, supporting real-time monitoring of financial markets, trading activity, order flow, options flow, commodities, cryptocurrencies, and live market data.
- Developing APIs and backend services for real-time market monitoring, financial data processing, market layers, trading workflows, and analytics.
- Building asynchronous and event-driven processing systems using AWS SQS, Redis, RabbitMQ, Apache Kafka, and ActiveMQ for distributed processing and background workloads.
- Building ETL and data-processing pipelines to transform and structure large volumes of financial data for analytics, monitoring, and AI-driven applications.
- Integrating AI and LLM technologies, including OpenAI APIs, Anthropic, Ollama, and other AI models, into backend workflows for intelligent data analysis and natural-language interaction with structured data.
- Integrating AI assistants with Google BigQuery to enable natural-language querying and analysis of structured financial datasets.
- Contributing to AXIVIS, an AI-powered medical platform, developing backend services and AI-enabled healthcare workflows for patient data, laboratory reports, medical information processing, and personalized health intelligence.
- Implementing asynchronous AXIVIS workflows using AWS SQS, Redis, WebSockets, and background workers for long-running AI and healthcare data-processing tasks.
- Working with medical AI technologies including MedGemma, OpenAI APIs, Anthropic, and Ollama to support AI-assisted medical data analysis and healthcare workflows.
- Working extensively with AWS cloud services, including Elastic Beanstalk, ECS, SQS, CloudFront, S3, and Secrets Manager, for application deployment, asynchronous processing, configuration, and production infrastructure.
- Working with Google Cloud Platform, particularly BigQuery and data-streaming services, for large-scale data processing, analytics, and AI applications.
- Managing containerized and distributed applications using Docker and Kubernetes across development, staging, and production environments.
- Implementing CI/CD and engineering workflows using GitHub, automated testing, database migrations, deployment pipelines, health checks, and staging validation.

- Troubleshooting and maintaining cloud-hosted applications across development, staging, and production environments, including AWS infrastructure, application configuration, databases, Redis, message queues, APIs, and deployment issues.
- Collaborating across backend engineering, AI/ML, cloud infrastructure, DevOps, and data engineering to design, implement, test, deploy, and maintain production systems.

Nanjing University of Information Science and Technology – Nanjing, China | Research Analyst in AI | Jan 2025 – July 2025

- Conducted research on AI-based medical imaging analysis, focusing on histopathology and prostate cancer grading.
- Worked with deep learning models such as UNet and SegGini for weakly supervised semantic segmentation of whole-slide images.
- reprocessed large-scale histology datasets (.mrxs format) and implemented pipeline components for tile selection, annotation, and training.
- Evaluated segmentation and classification performance using metrics like Dice score, accuracy, and per-class analysis.
- Integrated RabbitMQ, ActiveMQ, and Apache Kafka for messaging, queuing, and data streaming solutions.
- Collaborated with supervisors and domain experts to integrate AI techniques into clinical research workflows.

Axiata Digital Labs - Sri Lanka | Senior Software Engineer | Apr 2023 - Nov 2024

- Developed and maintained telecom applications including NCell DMS (Distribution Management System), NCell PG (Payment Gateway), and Dialog StarPoints, by designing and implementing Java Spring Boot-based backend services using microservices architecture and deploying them as cloud-hosted REST APIs to ensure scalability and high availability. Developed applications using Java Spring Boot within a cloud-hosted, web-based microservices architecture.
- Worked with multiple databases including Oracle SQL, PostgreSQL, and Microsoft SQL Server.
- Deployed services in a Kubernetes and Docker cluster environment, enabling efficient containerized application management.
- Utilized Kibana and Grafana for real-time system monitoring, log analysis, and debugging of production issues.
- Integrated RabbitMQ, ActiveMQ, and Apache Kafka for messaging, queuing, and data streaming solutions.
- Analyzed, implemented, and deployed new Customer Requirements (CRs) across multiple systems, ensuring alignment with business needs, seamless integration with existing services, and minimal disruption to production environments.
- Provided Level 3 support for business-critical telecom and fintech applications, ensuring minimal downtime and optimal performance.
- Managed customer interactions by addressing technical concerns, gathering feedback, and resolving issues in line with SLAs.
- Conducted training sessions for team members on system usage and troubleshooting to ensure smooth transitions and maximize operational efficiency.
- Analyzed distribution metrics to drive continuous improvement and enhance overall customer experience.
- Delivered Knowledge Transfer (KT) sessions to share system overviews and ensure smooth onboarding for new team members

Axiata Digital Labs - Sri Lanka | Software Engineer | Aug 2021 - Mar 2023

- Contributed to the development of NCell Payment Gateway (PG) and Dialog DMS (Distribution Management System) as part of the backend and integration teams.
- Gained hands-on experience with Java Spring Boot, including Spring Security, Spring Data JPA, Spring MVC, and Spring Cloud, through self-learning and project work.
- Worked on frontend development using Angular, contributing to feature implementation and UI enhancements.
- Developed modules using ASP.NET and ASP.NET Core within the Dialog DMS project, handling both backend logic and service integration.
- Handled database components such as stored procedures, views, and triggers across Oracle, PostgreSQL, and SQL Server environments.
- Managed server deployments and configurations in Linux environments, including application hosting, log monitoring, and basic shell scripting.
- Participated in smaller-scale projects using Python, contributing to automation scripts and backend service tasks.
- Provided timely responses to business support requests and user inquiries, delivering support in alignment with established SLAs.

Apr 2021 - Aug 2021

- Developed and maintained the Performance Monitoring and Fault Management (PMFM) System using the PHP Laravel framework, implementing backend functionalities including CRUD operations, email handling, file upload and storage, authentication management, and server configuration. Utilized MySQL for database management, Laravel's built-in email and auth features, and handled deployment and hosting.
- Developed the Foreigner Safety Management System, a PHP Laravel-based web application featuring CRUD operations and real-time tracking functionalities using location-based services. Leveraged MySQL and MongoDB for data management, integrated Google Location APIs, and handled server hosting and deployment.

Centre for Defence Research & Development, Ministry of Defence - Sri Lanka | Software Engineering Intern | Sep 2020 - Mar 2021

- Developed and developed a comprehensive Android application using Java and the Android SDK to analyze bullet impact images and accurately calculate the angle of incidence.
- Utilized SQLite for local data storage and built the application to support forensic investigations.
- Delivered to the Sri Lanka Police for field implementation upon successful development, testing, and validation.
- Publication (Elsevier): <https://authors.elsevier.com/c/1dVKI9UFWMsVTm>

Freelance Graphic Designer | 2017 - Present

- Worked with diverse local businesses, YouTube channels, and international clients on graphic design and video editing projects.
- Delivered creative solutions for branding, social media, promotional content, and visual storytelling.
- Proficient in Adobe Photoshop, Illustrator, Premiere Pro, and After Effects to produce high-quality visuals tailored to client needs.

PROJECTS

Carbon Footprint Management System

- Developed for JAT Holdings (Pvt) Ltd. using PHP Laravel, Bootstrap, jQuery, and MySQL.
- Designed to streamline data entry, calculation, and reporting of carbon emissions for organizational sustainability tracking.
- Key features included basic CRUD operations, automated report generation, and a dashboard interface for real-time monitoring and analysis.

AI-Powered Customer Service Chatbot

- Developed an AI-powered customer service chatbot using OpenAI models to handle customer queries and provide automated, context-aware responses.
- Implemented backend services using Python and FastAPI to process incoming messages, manage conversations, and integrate AI-powered natural language responses.
- Integrated Meta's WhatsApp Cloud API and Instagram Messaging API to enable automated customer interactions across WhatsApp and Instagram.
- Configured Meta Webhooks to receive and process incoming messages and route them through the chatbot backend.
- Integrated Supabase PostgreSQL for storing customer information, conversation history, chatbot sessions, and related application data.
- Designed RESTful APIs to connect messaging platforms, AI services, and database components into a unified chatbot system.
- Containerized the application using Docker and prepared the system for cloud deployment.
- Utilized Python, FastAPI, OpenAI APIs, Meta APIs, WhatsApp Cloud API, Instagram Messaging API, PostgreSQL/Supabase, REST APIs, Docker, and cloud deployment services.

#Find - Predict Social Media Hashtags

- Designed a deep learning and computer vision model combining CNN and LSTM architectures to classify video clips and predict relevant hashtags.
- Trained on a modified UCF101 dataset enriched with web-scraped hashtags, effectively capturing visual-temporal patterns for accurate tag generation.
- Applied hyperparameter tuning and regularization techniques to improve model performance and ensure better generalization across unseen data.
- Publication (ijsred): <http://www.ijsred.com/volume7-issue2-part16.html>

SKILLS

- **Programming Languages:** Python, Java, C#, JavaScript, TypeScript, PHP, SQL
- **Backend & Frameworks:** FastAPI, Django, Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Spring Cloud, Laravel, ASP.NET, ASP.NET Core, Node.js, Angular, React, Express, Thymeleaf
- **AI & Machine Learning:** OpenAI APIs, MedGemma, Anthropic, Ollama, TensorFlow, PyTorch, Keras, OpenCV, UNet, SegGini, scikit-learn, NumPy, Pandas, Matplotlib, Qdrant, Machine Learning, Deep Learning, Weakly Supervised Semantic Segmentation, Medical AI Workflows, LLM Integration, AI Agents & AI-Assisted Data Analysis
- **AI & Data Platforms:** BigQuery, Qdrant, AI-Powered Data Querying, LLM-Based Analysis, Medical AI Workflows, Financial Intelligence Systems, ETL/Data Pipelines, Natural-Language Data Querying
- **Databases:** PostgreSQL, Supabase PostgreSQL, MongoDB, MySQL, Oracle SQL, Microsoft SQL Server, SQLite, Firebase, BigQuery
- **Cloud & Infrastructure:** AWS (Elastic Beanstalk, ECS, SQS, CloudFront, S3, Secrets Manager), Google Cloud Platform (GCP), GCP Data Streams, GCP BigQuery, Cloud Deployment
- **Distributed Systems & Messaging:** Apache Kafka, RabbitMQ, ActiveMQ, Redis, AWS SQS, WebSockets, Event-Driven Architecture, GCP Streams
- **Architecture & APIs:** REST APIs, Microservices, Asynchronous Processing, Event-Driven Architecture, Distributed Systems, API Integration, SOAP, Meta Cloud APIs (WhatsApp Cloud API, Instagram Messaging API, Webhooks)
- **DevOps & Deployment:** Docker, Kubernetes, GitHub Actions, CI/CD, Jenkins, Cloud Deployment, Automated Testing, Database Migrations, Health Checks, Production Troubleshooting, ALDON, Windows System Administration
- **Monitoring & Observability:** Grafana, Kibana, Application Logs, Health/Readiness Checks, Level 3 Production Support
- **Testing & Development:** JUnit, Mockito, Postman, Swagger, Git, GitHub, GitLab, Forgejo
- **Build Tools:** Maven, Gradle
- **Operating Systems & Servers:** Linux, Windows, Apache Tomcat
- **Project & Delivery:** JIRA, Agile, Scrum, Code Review, CI/CD Workflows, Production Support, SLA Management, Customer Engagement, Knowledge Transfer (KT)

REFERENCES

References available upon request